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REVIEW DRAFT

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Subject: Cement Kiln Dust

In August 1999 the Environmental Protection Agency Office of Solid Waste proposed management standards for Cement Kiln Dust (CKD) waste. The following statements are taken from EPA's Environmental Fact Sheet issued at the time of the announcement of the proposed CKD management standards.

- Although current disposal practices cause some environmental damage, the Agency found that regulating cement kiln dust as a hazardous waste was not appropriate. Since some controls are needed, EPA is proposing a tailored set of standards for managing cement kiln dust waste.
- The Agency's preferred option is to provide management standards whereby CKD remains a nonhazardous waste so long as the waste is managed according to the requirements. Cement kiln dust becomes a regulated hazardous waste only if significant violations of the management standards occur.
- Under EPA's proposed standards, cement kiln dust is to be managed in landfills designed to meet specific performance requirements that protect ground water from toxic metals. In addition to performance criteria, the Agency is proposing technology-based standards that meet the performance criteria, such as using composite liners in landfills. Requirements for ground-water monitoring, corrective action, closure, and post-closure care are also included.
- To control releases of cement kiln dust to air, EPA is proposing a performance standard that requires facility owners and operators to take measures to prevent releases from landfills, handling conveyances, or storage areas. As an alternative to the performance-based standard, the Agency is proposing technology-based standards that require: (1) compacting and periodic wetting of CKD managed in landfills; (2) on-site handling of CKD in closed, covered vehicles and conveyance devices; and (3) keeping cement kiln dust in enclosed tanks, containers, and buildings when temporarily stored for disposal or sale.

There are three cement plants in Colorado. The Mined Land Reclamation Board holds permits on the limestone quarries associated with each of these plants as follows:

Holnam Inc., Portland Quarry, Florence, M-77-344.
Holnam Inc., Boettcher Quarry, Fort Collins, M-77-348

Southdown Inc., Lyons Quarry, Lyons, M-77-208

The cement industry has known that EPA would be proposing CKD management standards for some time. On August 5, 1998 a meeting was held with representatives from Holnam, the American Portland Cement Alliance, DMG and the Hazardous Materials and Waste Management Division in attendance. Holnam laid out their position on CKD management as follows:

- Protective management standards for the disposal of CKD are appropriate.
- In Colorado, the Hazardous Materials and Waste Management Division and/or DMG have authority to regulate CKD management.
- By exercising existing state authority, Federal rule may be unnecessary.
- Holnam wants to facilitate state agencies obtaining statutory authority to regulate CKD management.
- Holnam is seeking stringent regulation of management practices under state authority.

It was determined in discussions between DMG and Hazardous Materials and Waste Management Division that DMG is the appropriate agency to regulate CKD disposal. DMG regulation of CKD disposal is appropriate because:

- All three of the cement plants in Colorado dispose of CKD in mined out limestone quarry pits. CKD disposal in the quarries at all three sites has been ongoing for several years, and DMG has jurisdiction over the environmental impacts and reclamation of the quarries.
- DMG routinely permits and regulates the disposal of mining or processing waste generated from within a reclamation permit area. DMG does not generally regulate the landfilling of imported waste unless it meets the definition of inert waste in Rule 1.1(20). By bringing the areas of the cement plant where CKD is generated, stored or transported into the reclamation permit (see below), CKD becomes in effect a mining/mineral processing waste that DMG may regulate.
- DMG is the implementing agency for groundwater protection at active mines and thereby has the capacity to monitor for pollutants that may derive from CKD disposal.
- DMG has the authority to require stabilization of landfilled CKD, by periodic cover application or water application, to prevent dust generation (34-32.5-116(4)(j), C.R.S.).
- DMG has the authority to require final closure and reclamation of CKD landfills within the reclamation permit area.

The cement plants in Colorado are eager to come under DMG regulation of CKD disposal because CKD landfills not regulated in accordance EPA's proposed standards may be cited as unpermitted hazardous waste facilities. To this end, all three of Colorado's cement facilities have submitted technical revisions to incorporate CKD disposal standards into their reclamation permits. The technical revisions detail the geochemistry of the CKD, ground water protection measures and monitoring, dust control, and closure and reclamation. The procedure being followed by the DMG is to incorporate the CKD disposal standards into the permit through the technical revision process, then to include all areas of the operation where CKD is generated, temporarily stored, or transported into the permit through an amendment. From a regulatory standpoint, approval of the amendment makes the CKD an onsite rather than an imported waste. From a technical standpoint, the cement plant must be included in the reclamation permit area in order to provide regulatory controls over the generation of the CKD, e.g., the types of fuel or chemicals used in the kilning process, which may effect the geochemistry of the CKD. In particular, if a cement plant were to begin using alternative fuels such as wood, tires, or waste oil, the potential changes to the nature of the CKD should be evaluated.

This memo discusses some of the details of CKD generation and disposal at the three Colorado cement plants. A discussion of the geochemistry of cement and CKD are provided for background.

The Holnam-Boettcher operation and the Southdown-Lyons operation have completed waste characterization studies and have established groundwater monitoring programs. The Holnam-Portland operation has completed waste characterization and some groundwater monitoring, and the division is processing a TR for continued CKD disposal at the Portland plant. Based on (a) leach test results, (b) chemical analysis of pit water adjacent to one of the quarries, and (c) CKD waste handling commitments, groundwater monitoring did not seem justified at the Lyons operation. The latter point – "CKD waste handling commitments" – was most important in reaching that determination and is discussed in the following sections.

A PRIMER ON PRODUCTION, CHEMISTRY, AND PROPERTIES OF CEMENT

Cement is produced by burning limestone and clay at very high temperatures in an inclined rotary kiln. It can take up to 2 hours for the raw materials to pass through the kiln depending on its length. As the mixture moves down the cylinder, it progresses through four stages of transformation. Initially, any free water is driven off, then calcination occurs by driving off bound water and carbon dioxide. After calcination, the limestone has been converted to lime (CaO). The third stage is called clinkering, where lime and decrepitated clay combine to form calcium silicates and calcium aluminates as shown in the following equation. The fourth stage in the process is the cooling of the clinker.

$$\underset{\text{(limestone)}}{CaCO_3} + \underset{\text{(clay)}}{SiO_2 + Al_2O_3 + Fe_2O_3 + H_2O(\text{bound})} + \text{heat}$$

$$\rightarrow \underset{\text{(tricalcium silicate)}}{3CaO \bullet SiO_2} + \underset{\text{(dicalcium silicate)}}{2CaO \bullet SiO_2} + \underset{\text{(tricalcium aluminate)}}{3CaO \bullet Al_2O_3} + \underset{\text{(tetracalcium aluminoferrite)}}{4CaO \bullet Al_2O_3 \bullet Fe_2O_3}$$

The compounds shown on the product side of the equation comprise 90 percent of portland cement. When water is added the two calcium silicates, which form approximately 75 percent of cement by weight, react to produce tobermorite gel and calcium hydroxide. Tobermorite gel is the main cementing component of cement paste. The average diameter of a grain of portland cement as ground from the clinker is about 10 μm. The particles of the hydration product, tobermorite gel, are on the order of a thousandth that size. The enormous surface area of the gel (about 3 million cm^2/g) results in very large attractive forces, or cementation.

From the above discussion, it is clear that it is through the addition of water and hydration of cement that curing and hardening may occur. Concrete does not dry out to harden, as is commonly thought. When concrete dries, it actually stops getting stronger. The reaction of water with cement in concrete may continue for many years after the concrete is poured, and the strength of the concrete will continue to increase. Each of the basic components of portland cement contribute to its behavior. Upon the addition of water to cement, tricalcium silicate rapidly reacts to release calcium ions, hydroxide ions, and a large amount of heat. The pH quickly rises to over 12 because of the release of alkaline hydroxide (OH^-) ions. This reaction is primarily responsible for the high early strength of hydrated portland cement. Hydrated tricalcium silicate compound attains most of its strength in 7 days.

Dicalcium silicate takes several days to set. It is primarily responsible for the later-developing strength of portland cement paste. Since the hydration reaction proceeds slowly, the heat of hydration is low. Hydrated dicalcium silicate compound produces little strength until after 28 days. Tricalcium aluminate exhibits an instantaneous or flash set when hydrated. It is primarily responsible for the initial set of portland cement and gives off large amounts of heat upon hydration. Gypsum added to portland cement during grinding of the clinker combines with tricalcium aluminate to control the time to set. Hydrated tricalcium aluminate compound develops very little strength, and shows little strength increase after one day, but is useful in varying concentrations and in combination with gypsum to control set times. Fast setting cement, with high concentrations of tricalcium aluminate, is less resistant to sulfate attack. Tetracalcium aluminoferrite also hydrates rapidly and develops only a low strength, but it does not exhibit a flash set.

In addition to varying the composition of the four major components of cement discussed above, speed of hydration is also affected by:

- Fineness of grinding. To achieve faster hydration, cements are ground finer.
- Amount of water added. Presence of a sufficient amount of water will speed the reaction rate and enhance workability of concrete. All concrete is mixed with more water than is needed for the hydration reactions. This is done to produce flowing concrete that will develop adequate 7 and 28 day strength. However, water not consumed in the hydration reaction will remain in the microstructure pore space. These pores make the concrete weaker due to the lack of strength forming calcium silicate hydrate bonds. Thus a higher water:cement ratio yields a lower strength concrete, and workability and reaction rate must be balanced against ultimate strength.
- Higher temperatures of the constituents (cement, water, aggregate) at the time of mixing will also speed the hydration reaction rates.

Certain admixtures may be added to concrete that will modify its characteristics. Use of admixtures must be evaluated carefully since improvement of one characteristic often results in an adverse effect on another characteristic. In addition to admixtures, concrete properties may be varied by using the different grades of portland cement that are available and by adjusting the basic ratios of the concrete mixture of water, cement and aggregate.

- Air-entraining agents, usually natural or synthetic soaps, create microscopic bubbles within the paste to relieve pressure and prevent cracking during freezing conditions.
- Calcium chloride is used to accelerate the set and development of strength of concrete. It improves workability, reduces bleeding, and creates a more durable surface. Possible problems include increased drying shrinkage, increase in the rate of heat liberation, and corrosion of reinforcing steel.
- Pozzolanic materials are siliceous substances (e.g., fly ash or pumice) that can be used in combination with or for partial replacement of portland cement. Pozzolan enhances workability with less total water, provides cost reduction through cement savings, reduces the heat of hydration, and increases resistance to sulfates. Disadvantages of pozzolans include slow development of final strength, increased drying shrinkage, and impaired durability.
- Set retarders may increase setting times for concrete by several hours. These agents generally improve workability so the amount of mixing water may be reduced with an increase in ultimate compressive strength.
- Superplasticizers increase ultimate concrete strength by decreasing the amount of water needed to prepare workable concrete.
- Five types of portland cement have been classified by ASTM specification C150. Types I and II are general purpose with Type II having lower heat of hydration and enhanced sulfate resistance by limiting tricalcium silicate and tricalcium aluminate content. Type III is a high early strength cement prepared by increasing tricalcium silicate and tricalcium aluminate and by finer grinding. Type IV is a low-heat-of-hydration cement prepared by limiting tricalcium aluminate and tricalcium silicate. This limitation also results in lower early strength. Low heat of hydration is critical in mass-concrete applications because heat can expand the volume of a large pour while the concrete is plastic, then differential cooling after hardening causes shrinkage cracking. Type V is a sulfate resistant cement prepared by reducing sulfate susceptible tricalcium aluminate content to a minimum. Note that the pozzolan admixture can enhance Type IV and V cements, while at the same time reducing cost.
- The water cement ratio is the prime factor affecting the strength of concrete. In applications where high strength is critical, the mixing water should be reduced to the minimum needed to attain necessary workability.
- Cement content in the concrete mix is also critical to ultimate compressive strength, with strength decreasing as cement content is decreased. However, cement is by far the most expensive component of concrete, so an economic advantage is gained by using as little as possible while obtaining the minimum specified strength.
- Contrary to popular belief, concrete does not harden by drying, but rather by hydrating. Curing conditions are vital to the development of ultimate compressive strength, and moisture must be maintained in the concrete during the curing period. Concrete strength is detrimentally affected by curing in a dry atmosphere.
- Mixing water needs to be pure in order to prevent side reactions from occurring that may weaken the concrete or otherwise interfere with the hydration process. Potable water is often specified.

CEMENT KILN DUST

Cement kiln dust (CKD) is a by-product of cement manufacturing. It is an inorganic material collected in the air pollution control devices of portland cement manufacturing plants and is the finely-divided particulate matter carried from the cement kiln by exhaust gases. The dust is composed of variable mixtures of calcined and uncalcined feed materials, fuel combustion byproducts, condensed alkali compounds, and fine cement clinker formed during the high temperature processing. Alkalis may be concentrated in the dust through volatilization in the high temperature zones in the kiln then condensed in the exhaust gases as they pass through the dust collection system. The composition of CKD varies depending on production conditions and the nature of the raw material and fuel. The actual form of the components may typically be:

Calcium Carbonate ($CaCO_3$)	10 percent
Available Lime (as CaO)	30-50 percent
Potash (as Na and K salts)	6-10 percent
Chloride	4 percent

Remembering the stages of cement clinkering in the kiln, it can be seen that CKD example above is made up of the raw materials for clinker manufacture, which are limestone ($CaCO_3$) and calcined limestone or lime (CaO). It makes sense that the kiln exhaust would contain dust made up of pre clinker raw materials, because by the time clinker is formed the material is molten and would not produce much dust. Most cement plants recycle their kiln dust into the kiln. However, complete recycling is not possible due to a build up of undesirable elements as discussed below. CKD is not portland cement and does not exhibit the complex cementitious properties of portland cement. For most CKD, the chemical and physical behavior is dominated by the lime (CaO) component. CKD contains little calcium silicate, which is the basic ingredient of cement. Continuous recycling of CKD concentrates alkalis (potassium and sodium), chloride, and sulfate, which adversely affect the clinkering process and force the kiln operator to shunt a certain percentage of CKD. Recycled CKD also concentrates volatile metals making recycled CKD disposal potentially more likely to release metals to the environment.

This memo distinguishes between what is termed "fresh" CKD and "weathered" CKD. These materials differ in their physical properties and in their response to leach testing, metals mobility, and effects on pH. These differences are discussed. Some CKD may contain large quantities of calcined feed materials, alkalis and sulfur compounds, or both, while others may be primarily composed of uncalcined raw feed. Many CKDs are easily compacted and reactive with small quantities of water to form a low strength cementitious mass; some may show only mild reactivity and be non-consolidating. The following discussion of "fresh" and "weathered" CKD assumes a limey CKD.

"Weathered" CKD. As discussed above, most CKD is largely composed of lime. Lime is a simple cementing material produced by driving water and carbon dioxide off limestone (calcining). Its cementing properties arise from the reabsorption of the liquid and gas that has been expelled and the formation of chemical compounds similar to the original limestone raw material. Left exposed to the atmosphere, CKD will react, cure and harden. Its ultimate composition will become *limestone*. Thus, in that regard, *weathered CKD is not CKD* at all. It is limestone.

What will actually form when CKD is exposed to the atmosphere is a cohesive mass that looks like rock but which is soft, easily broken, and which can be pulverized with gentle abrasion even in the hands. In this memo, this type of CKD is called "weathered" CKD. "Weathered" CKD is distinguished from "fresh" CKD, described following.

"Fresh" CKD. The lime in "fresh" CKD – CKD that has not been exposed to the atmosphere – has a great affinity for water. When CKD is exposed to atmospheric moisture or meteoric water it begins to hydrate or slake. To fully slake CKD, it would have to be thoroughly mixed with two to three times it weight of water. This is unlikely to occur in most CKD disposal areas, but some portion of disposed CKD will slake. More so if the CKD is disposed of in a water filled pit. When the lime in CKD combines with water calcium hydroxide is formed in an exothermic reaction. The resulting product is a finely divided calcium hydroxide which, upon cooling, stiffens to a putty and will eventually season and cure to weathered CKD. Incomplete slaking of CKD, which is the likely situation at a quarry disposal site, will likely result in a variable mass of weathered and fresh CKD and hydrated CKD present as a fine powder. Unless there is significant dilution, CKD in water will increase pH. Paste pH measurements can range up to 12.

Source of CKD Pollutants. Fully "weathered" CKD, which is actually limestone, tends strongly not to release metals when placed in water. However, if pulverized, metals can be released, but pH does not increase. Release of metals due to pulverization is a surface area effect, alone; any rock, if crushed or ground will release more metal to solution in a fixed period of time than that same rock will release if left uncrushed. Because pulverization generates a high surface area, some "weathered" CKD can release high metals.

Elements that were present in the clay or limestone prior to kilning for cement manufacture may dissolve when "fresh" CKD is placed in water. The major element and trace element compositions of the clay and limestone prior to calcining determines what elements may appear dissolved in the water to which "fresh" CKD is added. Most of the dissolved metals come from clay, not limestone. As discussed previously, volatile metals tend to concentrate in CKD at kilns that employ continuous CKD recycling.

Limestone ($CaCO_3$) typically does not contain significant concentrations of trace elements other than strontium because the calcite structure accommodates elements only of specific charge and size. Locally, Zn and/or Ba can be high in some limestones, but most other trace elements appear at low levels in limestone. Moreover, limestones that are selected for cement manufacture are chosen for their low percentage of impurities (clay and other silicates) so tend toward more pure, or high $CaCO_3$ rocks. Thus, the limestone raw material component cannot contribute much dissolved metal when lime or "fresh" CKD is exposed to water.

Clay minerals, however, are noted for containing high concentrations of many trace elements. Informally, clay minerals are called "garbage" minerals because their structure accommodates elements covering the complete range of size and charge. While these elements are tightly bound and not significantly released during weathering, the process of calcining breaks bonds and disrupts the mineral structure, freeing up both the major elements and associated trace elements from the clay minerals. Thus, elements that may be released when CKD is exposed to water come dominantly from the calcined clay minerals. It follows that the pollutants which may derive from CKD in water will differ depending on the type of shale and its particular suite of clay minerals.

LEACH TESTING

Some of the above characteristics of CKD have become evident in part through leach testing conducted by the companies involved. Leach tests conducted on "fresh" CKD differ extremely from "weathered" CKD. The results are characteristic not only of the "type" of CKD but also the leach method.

"Weathered" CKD leach tests yielded small amounts of metals by most leach methods, but never high pH. On the other hand, natural water samples collected below "weathered" CKD piles tended not to show elevated metals or anomalous pH.

"Fresh" CKD leach tests produced elevated concentrations of several metals, regardless of the leach test method. "Fresh" CKD also produced high paste pH and leachate solution pH.

CKD HANDLING

From these tests, it is evident that CKD chemistry is controlled by weathering. Most fresh CKD bag-house dust is very reactive if placed directly in water. However, fresh CKD placed in thin layers on the ground where precipitation can reach will typically harden and cure to become a layer of pulverulent material similar to plaster or whitewash and consisting largely of calcium carbonate (limestone). If a CKD disposal plan calls for spreading in thin layers within a disposal cell, the CKD should be wetted to prevent fugitive dust emissions during spreading operations.

Disposal of CKD into water would not allow it to harden or cure. Disposal in water would result in elevated pH and may facilitate leaching of metals. Disposal below the natural ground water table should not be undertaken. Disposal into quarry pits filled with surface water should only be done after adequate evaluation of the potential for leachate migration and installation of down gradient ground water monitoring. The fact that a quarry pit fills with surface water is a good indication that the quarry is cut into low permeability strata, and the leachate migration will be minimal. CKD disposal should not occur in a floodplain unless it is demonstrated that the flood would not wash CKD into the stream.

CKD DISPOSAL

Based on field examinations of the three CKD sites, leach test results, and surface and groundwater analyses, the following general recommendations for CKD disposal are suggested. These suggestions distinguish between "fresh" CKD and "weathered" CKD – "fresh" CKD being that which is collected directly from a bag-house, and "weathered" CKD being that which has been in contact with the atmosphere, rain or snow for several weeks.

1. Fresh CKD should not be placed directly in ponded water. If conditions require such disposal, or if such disposal occurs inadvertently, the Division should institute groundwater monitoring.

2. "Fresh" CKD should never be placed, or allowed to be placed inadvertently, in open flowing water or in areas, such as flood plains, where open flowing water may reach the "fresh" CKD prior to a few weeks of curing.

3. "Fresh" CKD should be spread in thin layers (no more than a few feet thick) and immediately watered. Watering will help control dust and will aid the transformation from CKD to limestone. (Watering is effectively accomplished at the Holnam Portland Quarry in Florence where fresh CKD is covered with sewer sludge.)

4. Except where "weathered" CKD is to be subsequently covered and thereby disturbed, "weathered" CKD should not be significantly disturbed so as to re-pulverize it.

GROUNDWATER MONITORING PARAMETERS

Groundwater monitoring generally seems unnecessary unless "fresh" CKD is to be exposed to surface waters or unless "weathered" CKD is to be pulverized and exposed to groundwater.

1. Where groundwater monitoring is deemed appropriate, it is recommended that a representative suite of indicator elements be developed from site specific CKD leach tests and ambient groundwater conditions. Because clay compositions differ from site to site, a single set of parameters does not seem appropriate.

2. Monitoring parameters should ideally consist of elements or compounds that consistently reported above detection limits in groundwater samples or that consistently reporte well above detection limits in CKD leach tests.

3. Where sewer sludge, compost, or other organic materials or amendments are anticipated to come into contact with the CKD, or where reducing conditions can be expected to result, Mn might be used as a monitoring parameter. Mn is a typical component of limestone and can be mobilized under reducing conditions. Use of Mn for monitoring might be given extra consideration

4. Even though high alkalinity is a feature of "fresh" CKD weathering, it is not recommended that pH alone be used as a monitoring parameter where pollution from high pH might be anticipated. Because pH is a logarithmic representation of hydrogen ion activity, a unit shift in pH represents a 10-fold increase (or decrease) in H^+ activity. For instance, a pH 8 solution contains 1000 times greater hydrogen ion activity than a pH 11 solution; thus, unless the pH 11 solution comprises an extremely high percentage of a pH 8 and pH 11 solution mixture, pollution by the high pH solution would be masked and not detectable. A 50:50 mixture could not be detected.

5. Until the degree of variability on the dust source is well established, period leach testing of the CKD should be conducted. The technical revision for the Holnam Boettcher Plant requires TCLP testing on CKD samples semi-annually.

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